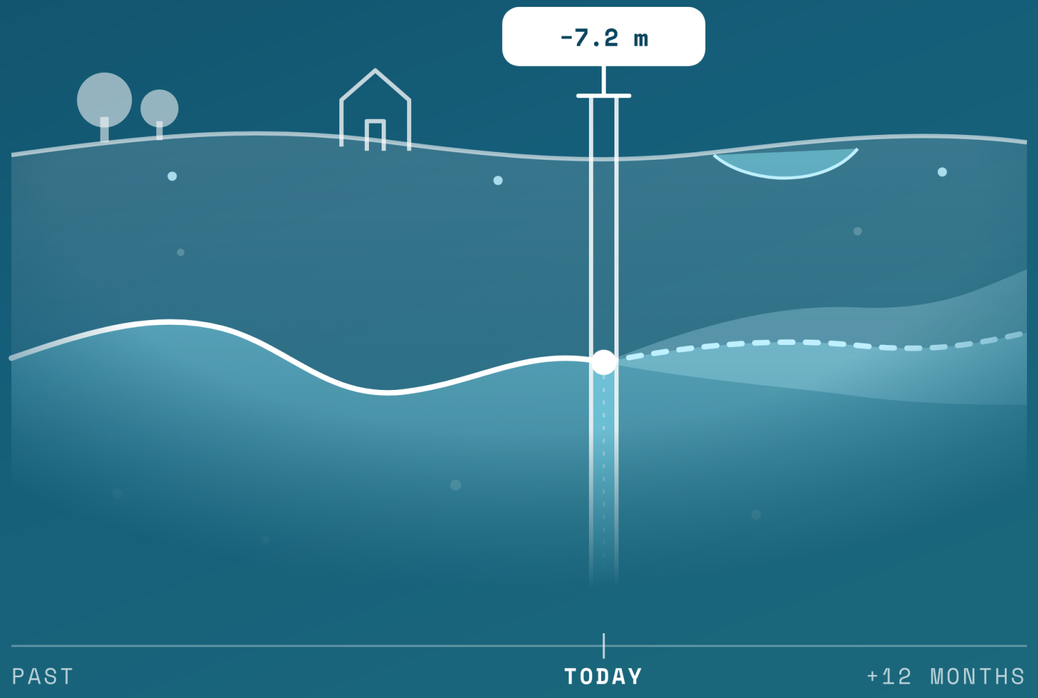
WATER RISK FORECASTING · OCTOBER 2026

PhreAItic

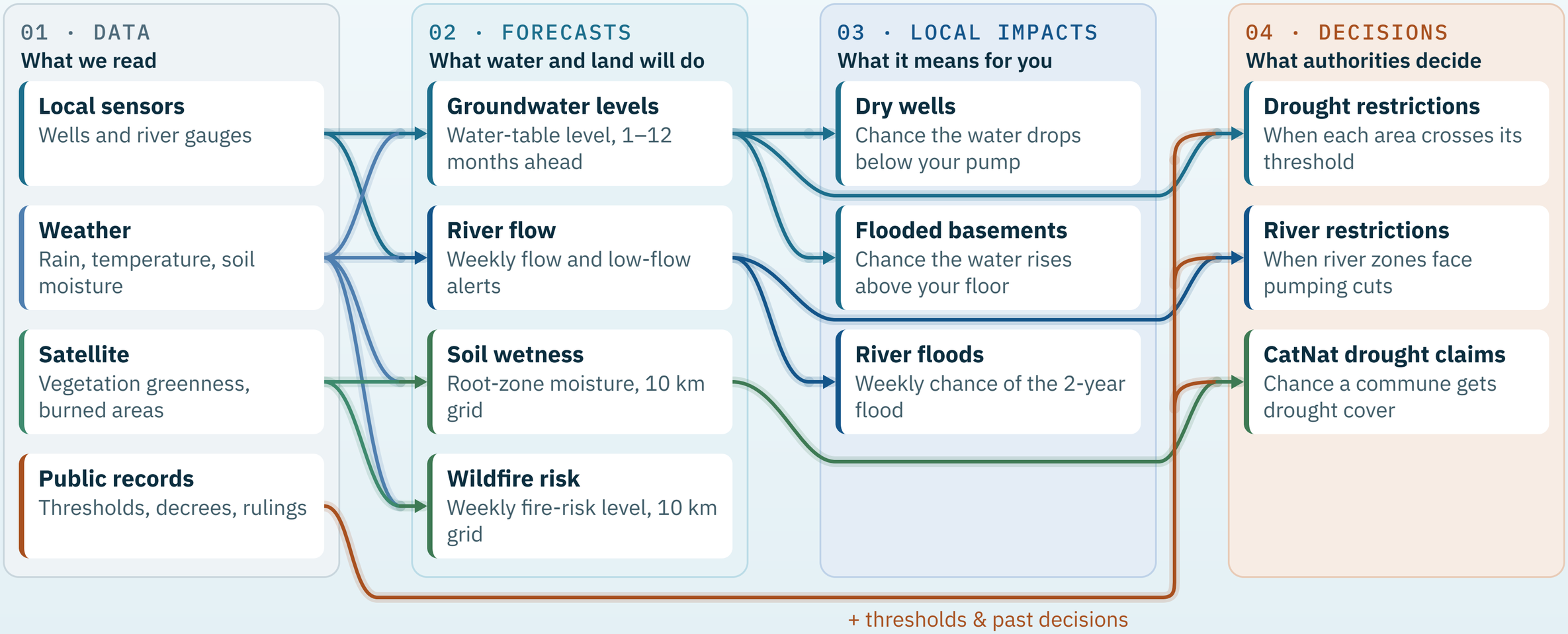
Groundwater, rivers, floods, soils, wildfire and drought rules: forecast weeks to a year ahead, with published accuracy.



HOW IT WORKS

PhreAItic

From public data to the decisions water triggers, forecast weeks to a year ahead



WATER · 01 Depends on local sensor data

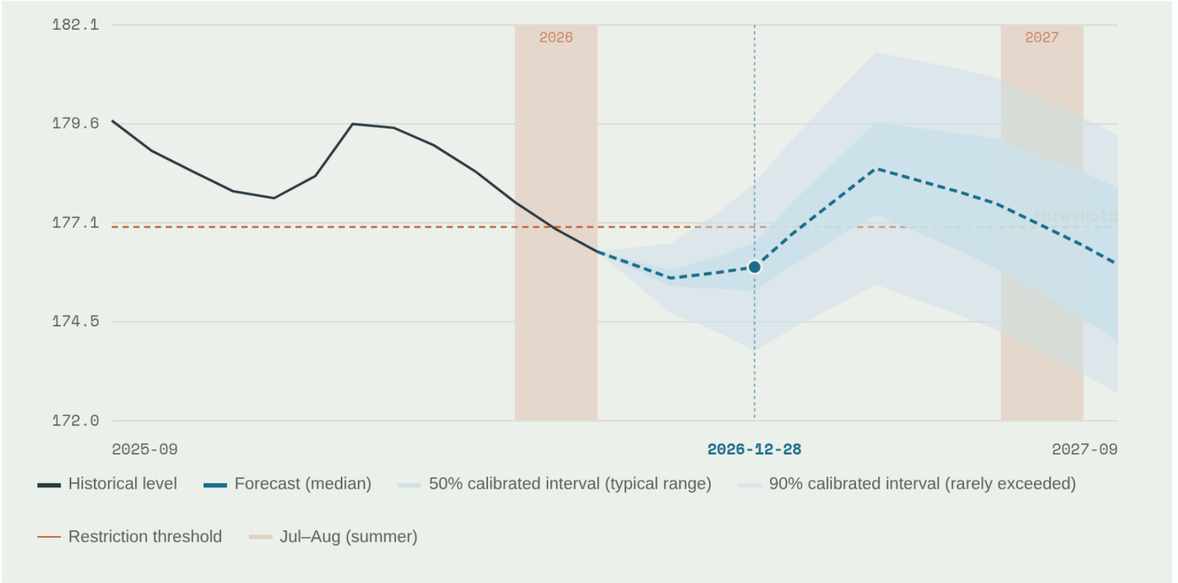
Groundwater levels

The water-table level at each monitoring well in France, 1 to 12 months ahead, with a calibrated range.

0.80

median skill one month ahead (1 = perfect, 0 = seasonal average)

Closer to the measured level than the seasonal average in 74% of 1-month forecasts. 90% range held 89% of the time up to 6 months.



Live forecast, Somme chalk aquifer: below its restriction threshold today, expected to recover over winter.

2,327 wells graded at 1 month. Weaker in fast-draining karst aquifers. Inputs: local water-level sensors and weather data.

WATER · DERIVED · 02 **Depends on local sensor data**

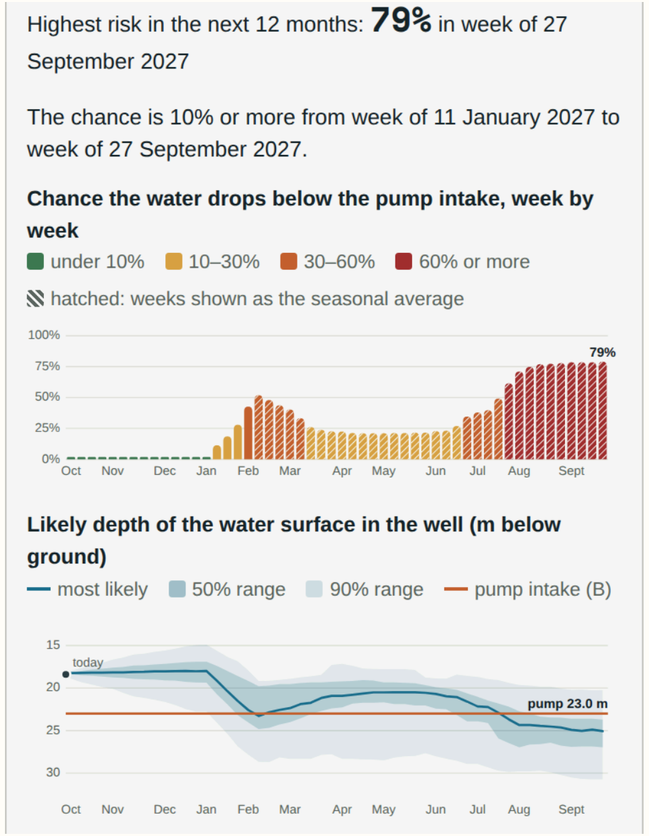
Dry wells

Enter a location and a pump depth: get the chance the water drops below the pump, every week for a year.

0.63 m

typical error three months ahead, vs 0.90 m for the seasonal average and 1.09 m for “no change”

50% and 90% ranges held exactly, on stations they were not fitted on.



Hérault, pump at 23 m: risk peaks at 79% in late September.

Mean error in metres of water depth, 2,118 stations backtested quarterly since 2014. Inputs: local water-level sensors and weather data.

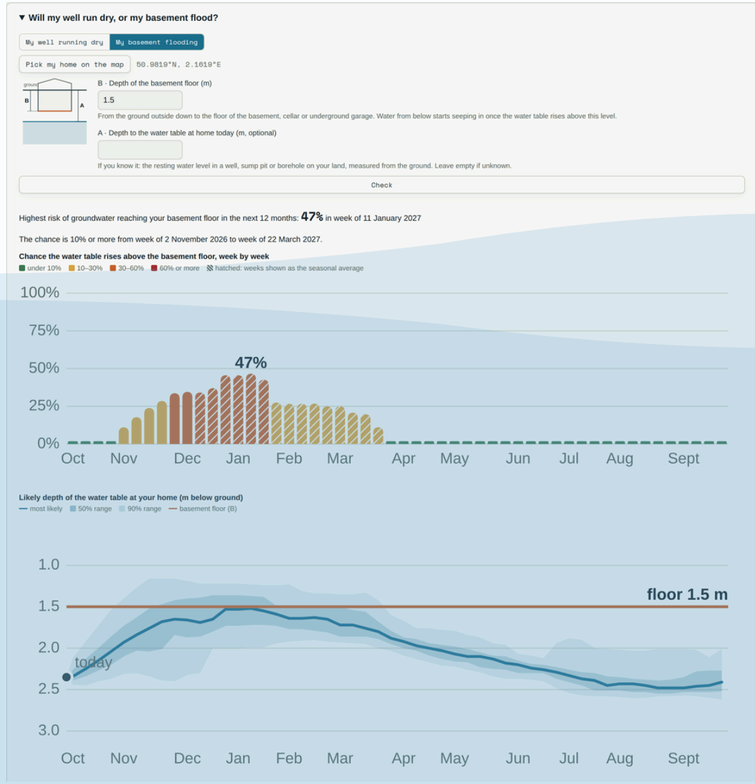
WATER · DERIVED · 03 **Depends on local sensor data**

Flooded basements

Enter a location and the depth of your basement floor: get the chance the water table rises above it, every week for a year. The map shows the same forecast at France's flood-prone monitoring wells.

> **54%**
>
> of weeks we rated “very high” danger really flooded, against 2% of “low” weeks

1–4 weeks ahead, we pick the flooding week over a dry one 92% of the time (seasonal record: 81%). 88% of official groundwater-flood decrees matched flood-level water at a nearby well.



Near Saint-Omer (Pas-de-Calais): a floor at 1.5 m peaks at 47% in January; at 2 m, over 95%.

Quarterly forecasts since 2014 at 278 flood-prone wells, ground level cross-checked against two elevation sources. Inputs: local water-level sensors and weather data.

WATER · 04 Depends on local sensor data

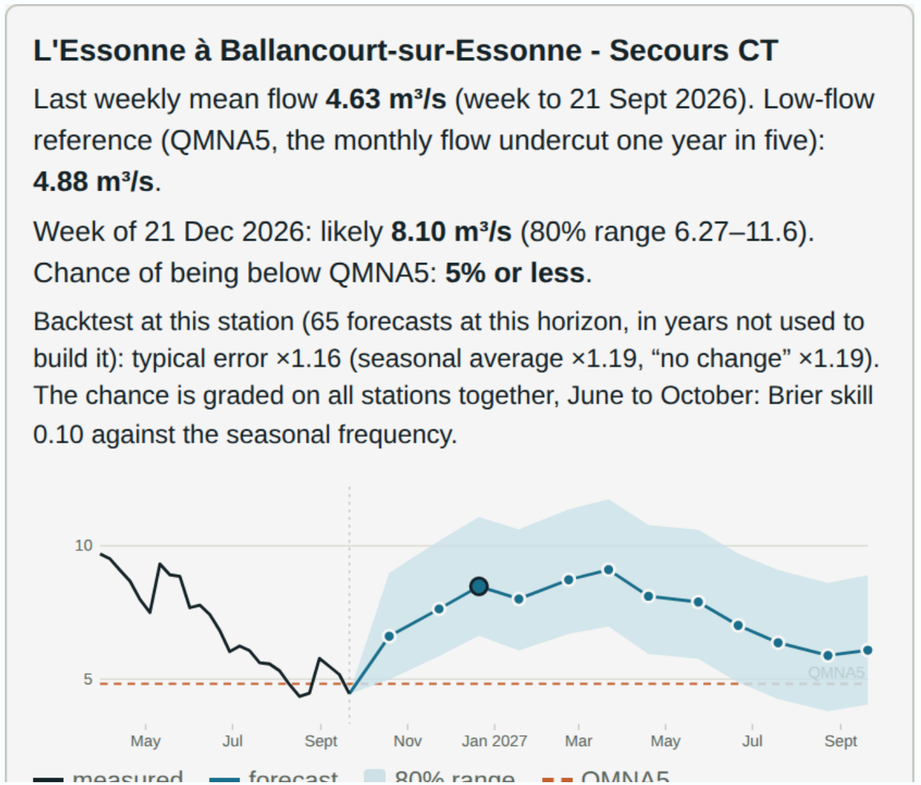
River flow

Weekly flow at river gauges across France, 1 to 12 months ahead, and the chance of falling below the low-flow reference.

×1.58

typical error one month ahead, vs ×1.71 for the seasonal average, at 81% of gauges

Closer to the real flow than the seasonal average in 60% of 1-month forecasts. Low-flow alerts beat seasonal odds at every horizon.



The Essonne, just below its low-flow reference today.

×1.58 = forecast usually within 58% of the real flow. Graded at 512 gauges on 2016–2025 years not used to build it. Inputs: local flow sensors and weather data.

WATER · DERIVED · 05 **Depends on local sensor data**

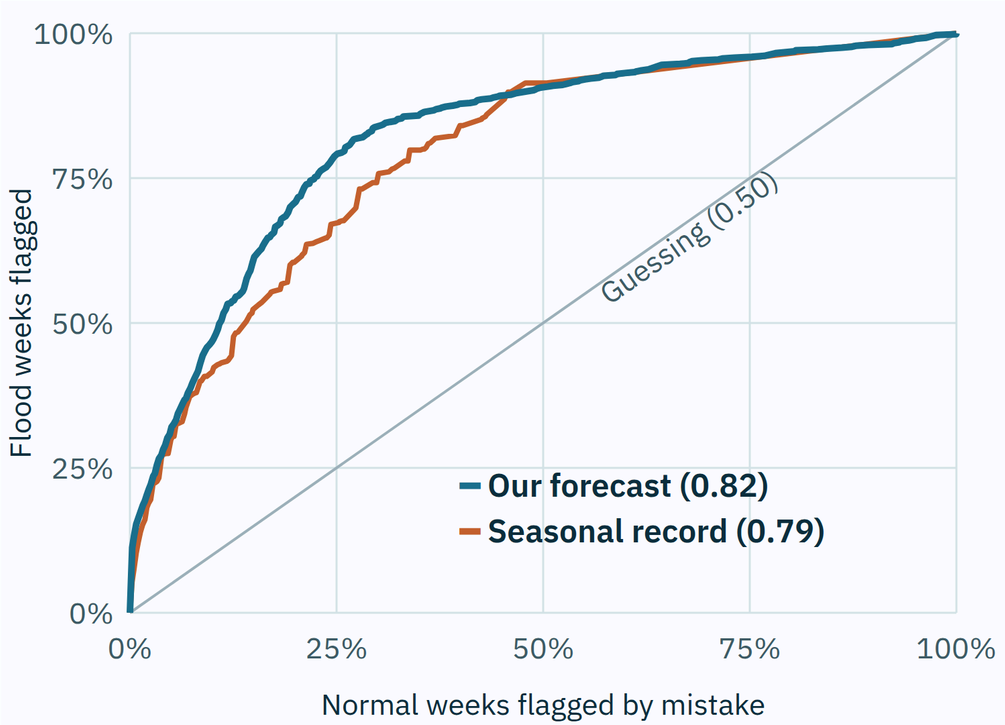
River floods

The chance a river reaches its 2-year flood (the daily flow exceeded one year in two, about where it leaves its bed) in a given week, at gauges across France.

60%

of weeks we gave a 30%+ flood chance really flooded, against 1.5% of weeks we gave under 5%

One month ahead it picks out flood weeks better than each gauge's seasonal record (ROC-AUC 0.82 vs 0.79). Shown only 1–2 months ahead: beyond that, the season says as much.



Each point is an alert level; the higher the curve, the more floods flagged for the same false alarms. Flagging 1 normal week in 10, we catch 47% of flood weeks; the seasonal record, 42%.

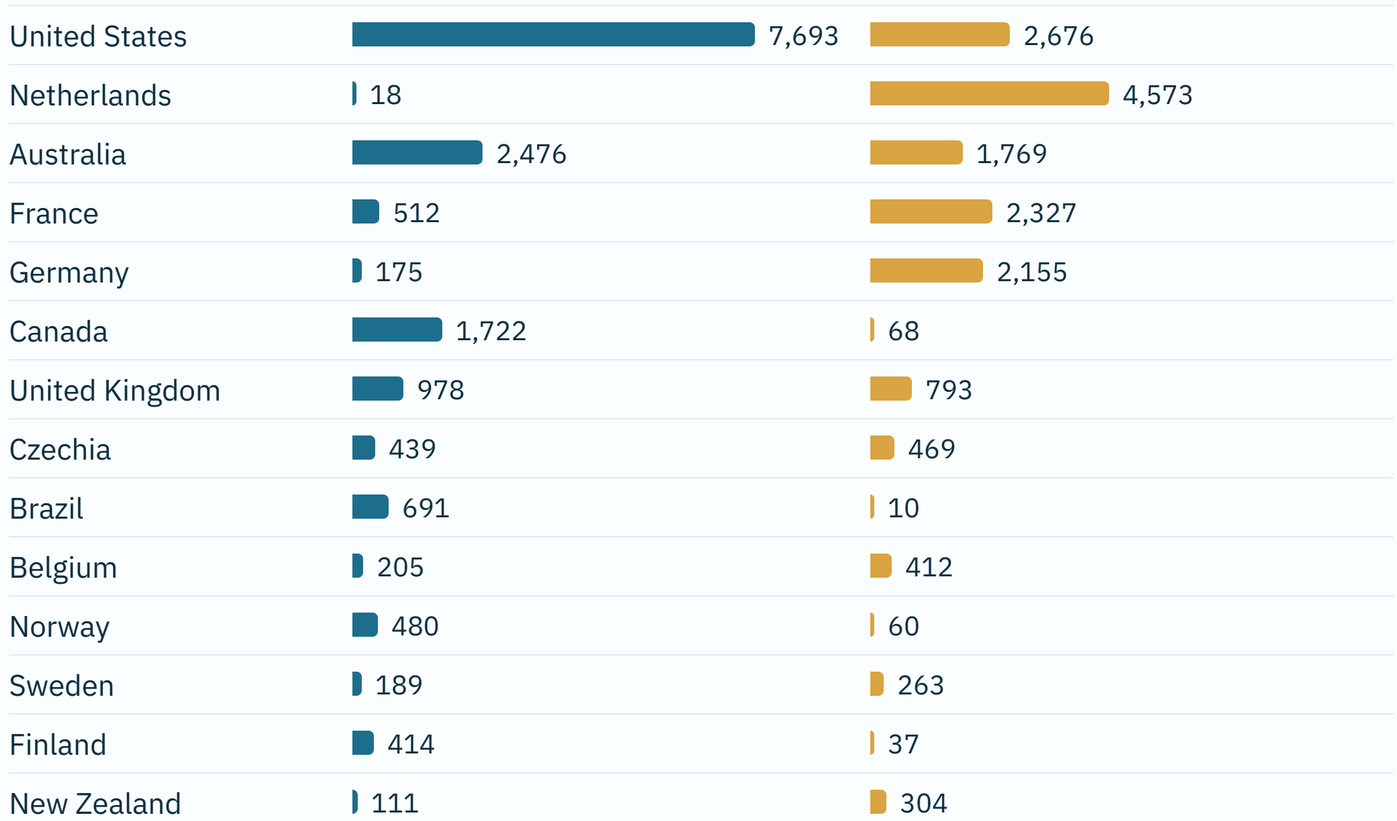
640 flood weeks at 414 gauges, one month ahead, graded on years not used to build it (2017–2026). Inputs: local flow sensors and weather data.

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WATER · GLOBAL COVERAGE Depends on local sensor data

River and groundwater forecasts, live in 43 countries

Stations forecast, 14 largest countries



35,385
river gauges and wells forecast 1 to 12 months ahead, 2,839 in France

57–96%
of river gauges where the forecast beat the seasonal average and “no change” one month ahead, by network

27–98%
of wells, likewise, by network

Next, in testing: data already in hand for further countries, waiting for their backtest; more agencies are checked every week.

Stations on the live map. Same method everywhere, graded on years not used to tune it.

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VEGETATION · 06 Available globally

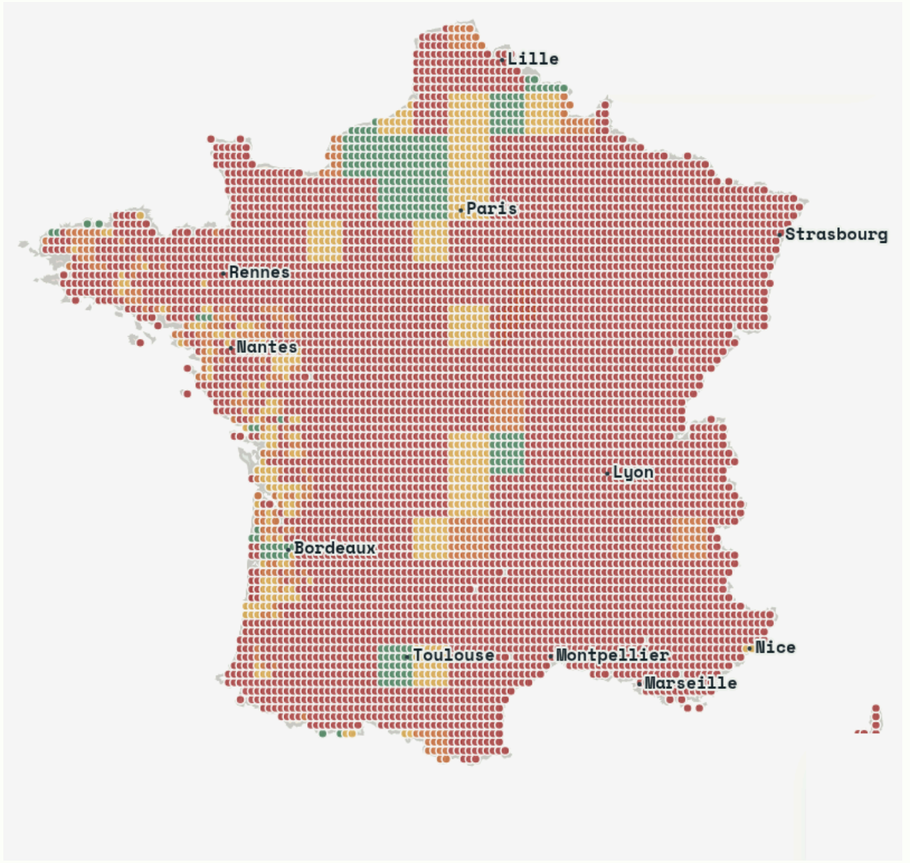
Soil moisture

How much water the soil holds 7–28 cm below the surface, where crops and grass draw most of their water. Ranked against the same week in past years, on a 10 km grid, 1 to 52 weeks ahead.

0.85

median skill one week ahead, and still 0.51 a year out

Beats “no change” in 99% of 60,104 checks.



Live reading: most of France in its driest range for the season.

Skill measured on held-out years at 1,158 backtest points. Inputs: satellite and weather data, available worldwide.

VEGETATION · 07 Available globally

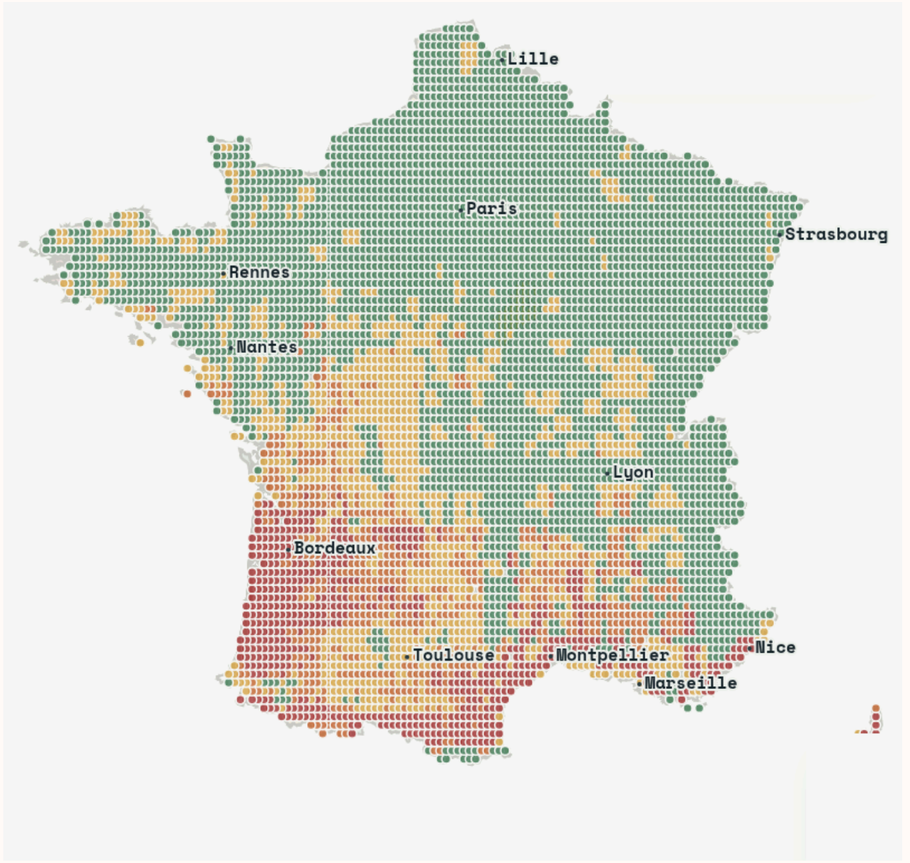
Wildfire risk

Weekly fire-risk level on a 10 km grid, with the fire rate observed at each level.

69%

of fires in test weeks fell in the 5% of cells we rated highest risk

ROC-AUC 0.97 on map cells left out of training. Research benchmark: about 2× the precision of Météo-France's daily bulletin, a week ahead instead of a day.



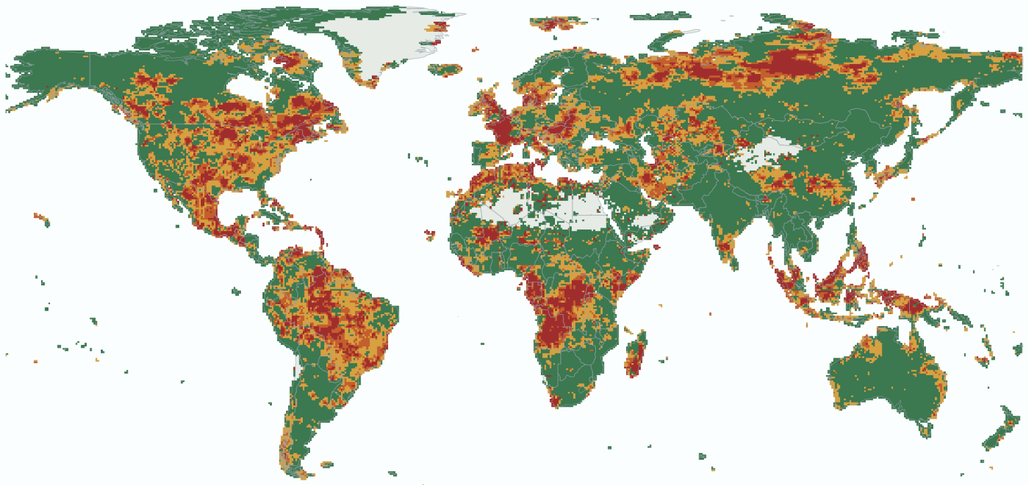
Live risk map: Landes forest and the Mediterranean rim stand out.

2,273 satellite-mapped fires in held-out weeks. Inputs: satellite and weather data, available worldwide.

PhreAItic

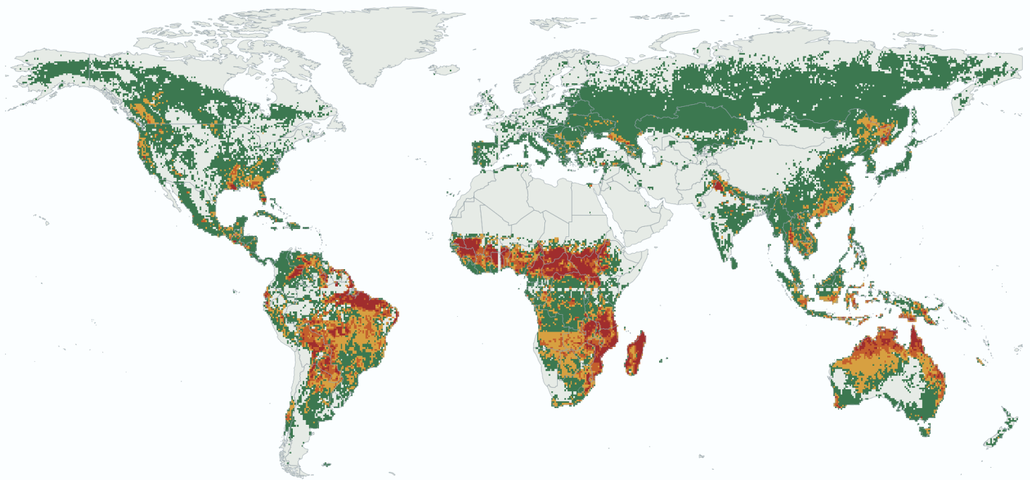
VEGETATION · GLOBAL COVERAGE **Available globally**

Soil moisture and wildfire, forecast in 230 countries



Soil moisture forecast for the week of 23 November 2026, compared with the same week in past years.

- Drier than 9 years in 10
- Drier than 8 years in 10
- Drier than 6 years in 10
- Usual or wetter



Chance of a fire month in November 2026, at the 33,102 places that have burned since 2001.

- < 10%
- 10–30%
- 30–60%
- ≥ 60%

54,227 cells forecast weekly, a year ahead

0.70 soil-moisture skill at 1 month

84% of “60%+” fire months burned

Graded per continent and shown only where it beats the simple reference. Inputs: satellite and weather data, nothing needed from local networks.

REGULATORY · 08 Depends on local sensor data

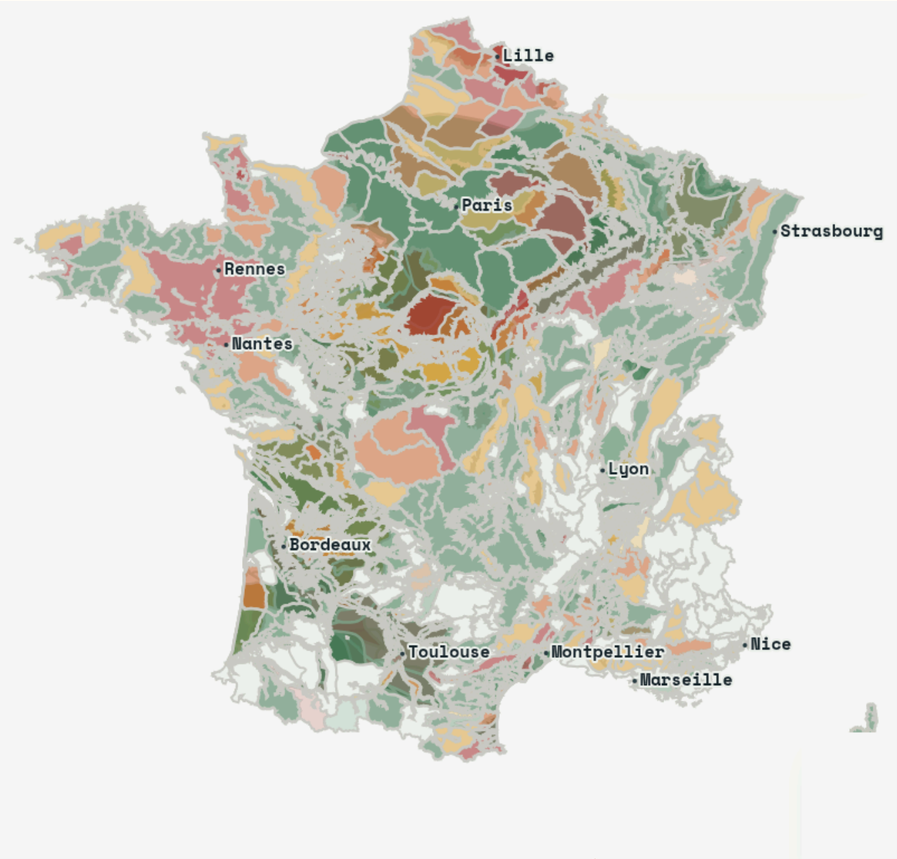
Drought restrictions

When each area will cross its official water-restriction threshold, and how severe it will be, from monitoring wells with a restriction history.

64 of 68

real threshold crossings caught one month ahead, timed to 2.6 weeks on average

A year ahead: 57 of 70 caught. In the Beauce, irrigation quotas forecast in October to within 0.06 of the April coefficient.



Forecast status three months ahead: green normal, yellow vigilance, orange and red alert levels.

Graded on past droughts at 1,274 stations. Inputs: local water-level sensors, official thresholds and weather data.

REGULATORY · 09 **Depends on local sensor data**

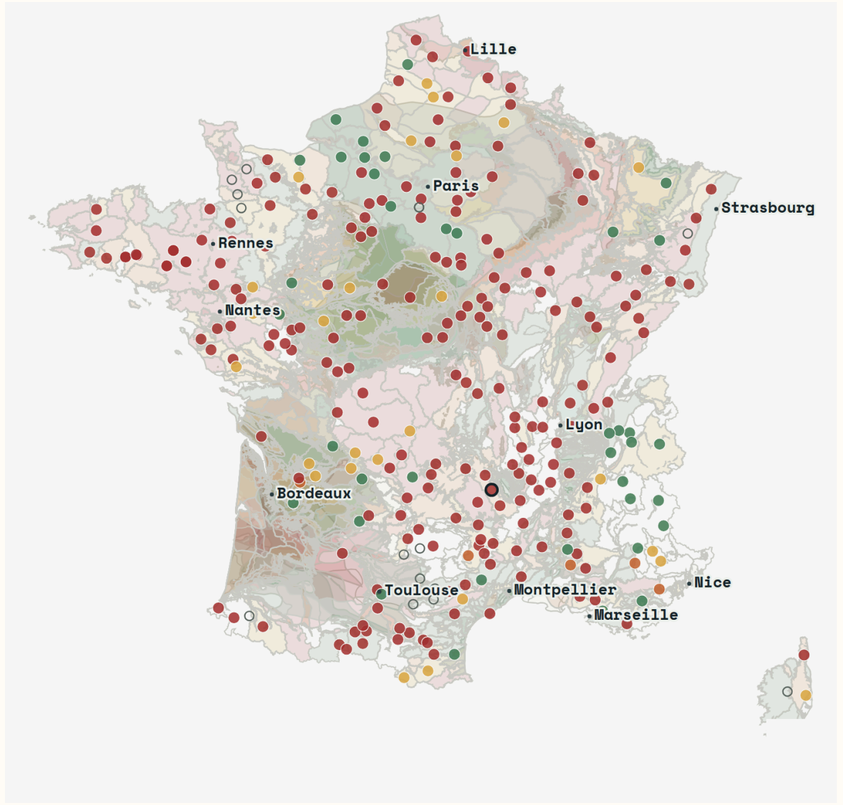
River pumping restrictions

The chance each river alert zone is under pumping restrictions (alert or worse), 1 to 6 months ahead, from our river forecasts and the official decree archive.

78%

of zone-months we gave a 60%+ chance one month ahead were really restricted, against 2.5% below 10%

Two thirds of new restrictions flagged a month ahead at 30% or more. Better than each zone's past record at every horizon: Brier skill 0.32 at 1 month, 0.12 at 6.



Live outlook one month ahead: 214 of 278 zones at 30% or more after a dry summer.

Graded 2019–2025 at 342 alert zones, using 512 river gauges, each year predicted from earlier years only. Inputs: local flow sensors, official decrees and weather data.

REGULATORY · 10 **Depends on local records**

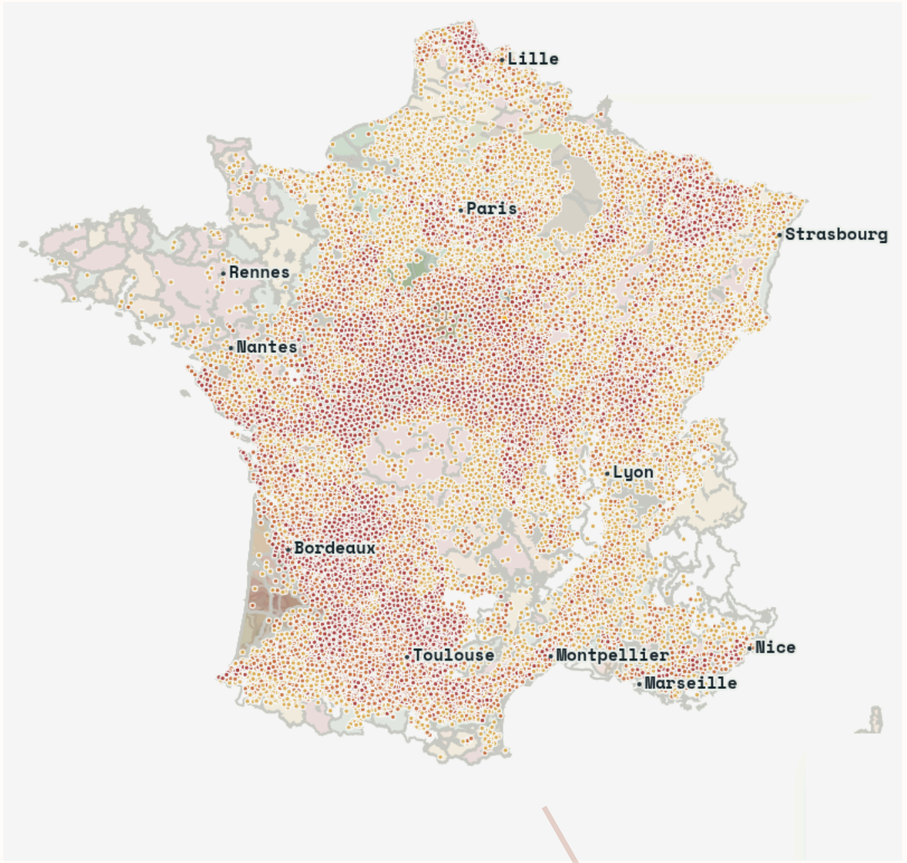
CatNat drought claims

The chance each French commune gets a CatNat drought recognition, which triggers insurance cover for cracked buildings.

0.87

ROC-AUC, vs 0.75 from each commune's past record alone

Better than the past record in 22 of 23 years. Issued end of September, months before the decrees.



2026 outlook: risk follows the clay belts.

Graded 2000–2022 on 34,746 communes, each year left out: 799,158 commune-years. Inputs: past insurance decrees, soil maps and weather data.

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REGULATORY · GLOBAL COVERAGE Depends on local data & rules

Official decisions, forecast in five countries

FRANCE	**Drought restrictions** groundwater thresholds · 1,274 wells	94%	of threshold crossings caught 1 month ahead (64 of 68)
FRANCE	**River pumping restrictions** VigiEau alert zones · 293 zones	0.90	ROC-AUC 1 month ahead
FRANCE	**CatNat drought claims** insurance recognitions · 34,746 communes	0.87	ROC-AUC (past record: 0.75)
UNITED STATES	**USDA drought-disaster trigger** US Drought Monitor rule · 3,222 counties	0.99	ROC-AUC 1 month ahead (season: 0.72)
CANADA	**BC drought levels** province drought level 4+ · 41 basins	0.90	ROC-AUC 1 month ahead (season: 0.61)
MEXICO	**Drought emergency** drought declarations · 2,477 municipalities	0.97	ROC-AUC 1 month ahead (season: 0.64)
TAIWAN	**Water restrictions** reservoir-based warnings · 17 water areas	0.95	ROC-AUC 1 month ahead (season: 0.72)

7
regulatory forecasts live, each graded on years not used to build it

5
countries today: each new one needs its own rules and decree archives

0.81
US drought-disaster ROC-AUC still six months ahead (season alone: 0.72)

What a new country needs: official thresholds, the archive of past decisions (restriction decrees, disaster declarations), and the sensors or drought maps they are based on.

ON EVERY GROUNDWATER-BODY PAGE

France only today · extendable with local data

Beyond the forecast: local context, in France today

Each item below comes from a French public registry. Any other country can get it too, as soon as it publishes the equivalent data: pumping declarations, hazard records, water-quality samples, restriction decrees.

Who pumps here

Declared withdrawals by use, year by year: drinking water, irrigation, industry.

Local risks

Official hazards on record nearby, such as groundwater flooding, plus the aquifer's type.

Water quality

Nitrate and pesticide levels and their trends since 2006, against drinking-water limits.

Past-year match

The past year this one most resembles, and what the water table did next.

Official status today

The restriction decree in force, linked to its text, and flood vigilance on the nearest river.

Why this forecast

Which current conditions push each forecast up or down.

On the map page of each of France's 470 forecast groundwater bodies (each item for 461–470 of them).

NEXT STEP

See it live today, then tell us where you need it

Where to find it

LIVE MAP, ALL FORECASTS

phreaitic.github.io

EVERY SCORE, PUBLISHED

.../trust_page.html



CONTACT

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A new area, on demand: what each forecast needs

Soil moisture, wildfire	Satellite and weather data: nothing needed from you
Groundwater, rivers, wells, floods	Local sensors (wells, river gauges), open or shared
Restrictions, quotas, claims	Local thresholds, allocation rules, insurance registries

Every new area is backtested first: a forecast goes live only where it beats the simple guesses there.